SAHAYAK | SCIENCE EXHIBITION
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LIVE DEMO, TESTING & CONCLUSION
ROLE-SPECIFIC SPEAKING SCRIPT
Suggested speaking time: 3-5 minutes

DEMO INTRODUCTION
Thank you. I will demonstrate how someone can try SAHAYAK and explain how we should judge whether it works reliably. This is a prototype, so the demonstration should show both its capabilities and its limitations.

LIVE DEMONSTRATION SCRIPT
I open the SAHAYAK website and go to Live Studio. I allow camera access and make sure my hand is visible in the frame. The landmark overlay shows the points detected by the browser. I perform a sign supported by the model, keeping my hand clearly visible. The extracted features are sent to the backend and the predicted label appears on the website. I repeat with another supported sign. If a prediction is wrong, I will acknowledge it and try again under clearer conditions rather than hide the error.

HOW TO TEST FAIRLY
One successful demonstration does not prove high accuracy. We should repeat trials for every sign, record correct and incorrect predictions, and test with people and recording sessions not used in training. We should vary lighting, distance, camera angle, and signing speed. Per-sign results and a confusion matrix can show which signs are confused. Response time should be measured separately from accuracy.

LIMITATIONS
The current model recognises a limited vocabulary. It may not understand full sentences, facial expressions, grammar, or every regional variation of ISL. Landmarks can be missed if hands are outside the frame, occluded, or poorly lit. Online translation and speech services may not work offline. This prototype is not a substitute for interpreters or a guaranteed communication service.

FUTURE IMPROVEMENTS
Future work includes collecting a larger, consent-based dataset with diverse signers, documenting data quality, testing on unseen people, measuring errors for every class, and improving sequence recognition. New signs should be added only after evaluation. Collaboration with ISL users and qualified experts is essential before wider use.

CONCLUSION
SAHAYAK demonstrates a pipeline combining camera input, hand-landmark detection, machine learning, and a web interface. It is a starting point rather than a finished universal translator. We have learned that assistive technology needs not only a working demo, but also representative data, careful testing, user feedback, and honest reporting. Thank you for listening. We are happy to answer your questions.

JUDGE Q&A

What if a prediction is wrong? We record the failure and investigate under controlled conditions.
Does it work offline? Not all features do; online backend inference and services require connectivity.
How can it improve? Collect diverse labelled data, prevent training/test data leakage, evaluate per class, and retrain only after verification.
How do you protect users? Request camera permission, minimise identifiable data, obtain consent for dataset collection, and explain what is transmitted or stored.

REMINDERS
Before judging, check internet, backend health, camera permission, and lighting. Keep a backup screenshot or architecture slide. Never invent accuracy figures. If you do not know an answer, say: We have not measured that yet, and it is one of our planned tests.